WiLo: Long-Range Cross-Technology Communication From Wi-Fi to LoRa

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***Abstract*—Wi-Fi is a very common means for providing wireless access to the Internet, e.g., using the 2.4GHz Industrial, Scientific, and Medical (ISM) band and more recently also the 6 GHz band via Wi-Fi 6E. Thanks to a chip recently launched by Semtech, in the same 2.4GHz band now can also operate Long Range (LoRa), which is widely used in Internet of Things (IoT) applications due to its low power consumption and wide coverage range. To allow for data interchange among these technologies, multi-radio gateways are needed, which introduce additional costs, complexities, and potential points of failure. To address this challenge, we propose the concept of Wireless to LoRa (WiLo) to make directional communication from Wi-Fi to LoRa. WiLo uses physical-layer (PHY) communication and dedicated input chips in the 2.4 GHz band to transmit information. To overcome the modulation technique differences between Wi-Fi and LoRa, WiLo leverages narrow-band communication, a technique that generates ultra-narrowband signals using single-tone sinusoidal signals by manipulating the payload of Wi-Fi devices. These signals can be detected by LoRa Wide Area Network base stations due to their high receiver sensitivity for long-range communication. Our experiments, which make use of both Universal Software Radio Peripheral (USRP) and commodity devices, demonstrate that WiLo can achieve concurrent wireless communication over a distance of 500 m, from commercial Wi-Fi chips to a LoRaWAN, with more than 96% frame reception rate. These findings show the effectiveness of WiLo in enabling reliable and efficient wireless communication over long distances, making it particularly relevant for applications such as remote monitoring systems, sensor networks, and smart cities.**

***Index Terms*—Cross-technology communication, LoRa, Wi-Fi, low-power wide-area networks.**

Received 30 May 2024; revised 11 August 2024; accepted 3 September 2024. Date of publication 16 September 2024; date of current version 19 March 2025. This work was supported by The NSF China 62102332. This work is part of the 6G-SENSES project from the Smart Networks and Services Joint Undertaking (SNS JU) under the European Union's Horizon Europe research and innovation programme under Grant Agreement No 101139282. The associate editor coordinating the review of this article and approving it for publication was G. Zheng. *(Corresponding author: Demin Gao.)*

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Digital Object Identifier 10.1109/TCOMM.2024.3461574

I. Introduction

THE exponential growth in the number of connected devices following each new wireless technology advancement [1], [2], particularly Low-Power Wide-Area Networks (LPWANs) and Wireless Personal Area Networks (WPANs), has been a significant trend in the past decade [3], [4]. The proliferation of IoT (Internet of Things) devices, which require reliable and efficient connectivity, has driven the development of these wireless technologies to enable seamless communication between devices [5], [6]. Wi-Fi has emerged as the predominant standard facilitating wireless connectivity to the Internet through the 2.4GHz ISM band [7]. An extensive proliferation of Wi-Fi-enabled devices globally has underscored its ubiquity, with Wi-Fi Access Points (APs) pervading public, enterprise, and personal spheres [8], [9]. Concurrently, to meet increasingly rigorous demands for board-space and power efficiency, numerous IoT devices are adopting wireless chips characterized by enhanced energy efficiency, reduced cost, and smaller form factors, features provided by technologies such as LoRa [10].

LoRa [11], a type of LPWAN, has been widely used in various IoT applications due to its long-range capabilities and low-power requirements [12]. Given the extensive adoption of Wi-Fi and LoRa, the necessity for data interchange among devices supporting different access technologies has arisen [13], [14]. Conventional approaches typically involve the utilization of multi-radio gateways to facilitate communication linkages between Wi-Fi and LoRa devices, which introduce additional costs, complexities, and potential points of failure. Notably, Semtech has introduced a new LoRa chip called SX1280 that operates in the 2.4GHz frequency band and offers a maximum available bandwidth of 1625 kHz, significantly higher than the previous 500 kHz, resulting in faster data rates [15]. This chip has emerged as a popular choice for wider support of IoT applications, particularly in low latency connections or remote locations where conventional networking solutions may not be available.

The adoption of the 2.4 GHz band for LoRa not only brings significant advantages but also presents an opportunity to harness the potential of Cross-Technology Communication (CTC) [16], [17] to further enhance the adaptation, popularity, and usage of Wi-Fi/LoRa in IoT applications [18], [19], [20]. CTC is a promising approach that allows for direct communication between different wireless technologies, enabling seamless interoperability without the need for additional multi-radio gateways [21], and re-purposing of existing wireless devices, overcoming challenges associated with traditional deployment methods, expanding the coverage and availability of Wi-Fi/LoRa connectivity, and enabling its deployment in previously inaccessible environments [22], [23].

In this article, we introduce a new algorithmic framework called WiLo, designed to enable directional communication from Wi-Fi to LoRa, which employs signal emulation techniques to enable off-the-shelf Wi-Fi hardware to produce a valid 2.4 GHz LoRa waveform. The system is designed to bridge the gap between traditional Wi-Fi and LoRa technologies by utilizing signal emulation, a technique that mimics the characteristics of LoRa modulation on standard Wi-Fi hardware. The core innovation of WiLo lies in the signal emulation technique used to generate a valid 2.4 GHz LoRa waveform. Through sophisticated signal processing algorithms, WiLo transforms the standard Wi-Fi signals into LoRa-like waveforms, while ensuring compliance with the LoRa modulation specifications. This enables the LoRa hardware to decode Wi-Fi signals without requiring any modifications to the hardware itself. The emulation of LoRa waveforms is achieved by carefully manipulating the parameters of the Wi-Fi signals, such as the modulation index, spreading factor, and BW, to closely match the characteristics of LoRa modulation.

The main contributions of this paper are summarized as follows:

- We introduce WiLo, a novel long-range signal emulation-based CTC scheme that enables communication from Wi-Fi to LoRa, which is designed to overcome the limitations of dedicated LoRa hardware by leveraging signal emulation techniques.
- WiLo has the ability to synthesize a valid LoRa frame within the constraints of the 802.11g physical layer, the standard for Wi-Fi communication. This allows for seamless integration of WiLo into existing Wi-Fi devices without requiring any changes to the hardware.
- We implemented the proposed system on commodity devices, specifically using the Universal Software Radio Peripheral (USRP) B210 and a LoRa evaluation kit from Semtech. Our evaluations demonstrate that WiLo enables CTC between Wi-Fi and LoRa with close to 98% frame reception rate.

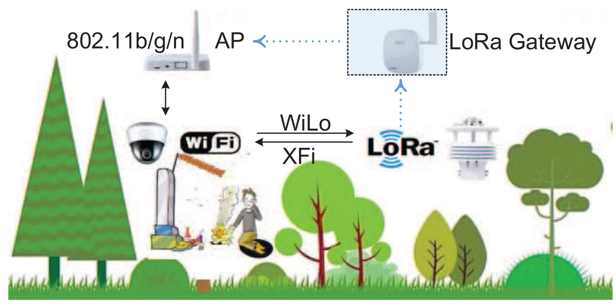


Fig. 1. LoRa and Wi-Fi systems are utilized for monitoring weather parameters and video frames taken by cameras.

The structure of this paper is as follows. Section II provides the motivation for our research. Section III gives an overview of needed preliminary information. Section IV describes WiLo and Section V details the proposed WiLo approach. Section VI provides performance evaluation results for WiLo and Section VII discuss the future research directions. In Section VIII, we review previous research on CTC for heterogeneous IoT systems. Finally, Section IX summarizes our conclusions and presents future directions for research.

II. Motivation

In this section we present the motivation behind our work, which is based on the following key observations.

A. WiLo for Forest Monitoring

Major forests monitoring and related information collection tasks are hard challenges due to the oftentimes not very accessible and widespread distribution of forests across the land. To overcome this, IoT devices such as Wi-Fi and LoRa can be used for collecting different types of data coming out of the deployed sensors: IEEE 802.11 b/g/n compliant Wi-Fi modules for transmitting the sensory data with high-rate technology, and LoRa devices, having a longer range, for the capability of forming one-hop networks with low-rate technology (called star networks). LoRa and Wi-Fi systems are utilized for monitoring weather parameters (temperature, humidity, wind speed, rainfall) and video frames taken by cameras, as shown in Fig. 1. Traditional methods for messages exchanging among heterogeneous devices rely on the concept of shared gateway, which always implies extra cost to be deployed (dashed box in Fig.1).

XFi [24] proposes a general approach to obtain IoT data by analyzing the decoded Wi-Fi payload for 802.11n/g when an IoT frame collides with an ongoing Wi-Fi transmission and its IoT data is captured by a Wi-Fi receiver. In other terms, directional communication from LoRa to Wi-Fi has been realized within the XFi framework. Therefore, enabling direction transmission from Wi-Fi to LoRa ensures a proper replacement of a LoRa gateway, and WiLo is purposed to meet such objective, albeit encountering certain challenges.

B. Challenge: The Extreme Asymmetric Data Rate of LoRa and Wi-Fi

In PHY-CTC technologies, using high-rate technology (e.g., Wi-Fi) to emulate low-rate technology (e.g., LoRa) is a formidable challenge, and asymmetric data rates can result in emulation failures in numerous scenarios. Currently, enabling Wi-Fi devices to establish connections and communicate with LPWANs based on LoRa technology, while maintaining high receiver sensitivity for long-range communication, is a straightforward concept but challenging to realize using existing CTC techniques. In particular, a high-rate packet can only emulate a limited number of symbols from a low-rate technology, resulting in a significantly shorter emulated payload compared to the original payload. This limitation poses challenges to the practical applications of CTC, and is especially severe in the case of CTC between Wi-Fi and LoRa, due to the substantial disparity in their data rates.

The extreme asymmetric data rate discrepancy often leads to emulation failures, with over 85% of common LoRa settings unable to be emulated by a single 802.11g frame, or even a complete LoRa symbol [25]. For instance, when considering the setting of SF=7 and BW=1625kHz, the duration of a single LoRa symbol can be as long as 79 μs, exceeding the duration of the longest 802.11g frame. The potential of utilizing the CTC from Wi-Fi to LoRa for the purpose of expanding wireless connectivity and enabling novel IoT application scenarios is compelling. However, the practical implementation of CTC from Wi-Fi to LoRa is impeded by its low spectrum efficiency, which poses challenges in real-world deployment scenarios.

C. Opportunity: Broad Deployment of Wi-Fi Devices and Longer Transmission Range of LoRa Devices

The maturity level achieved by CTC technology is particularly evident in the refinement of PHY waveform emulations, ranging from wide BW applications like Wi-Fi to narrower BW counterparts such as ZigBee and Bluetooth [26], [27], [28]. These advancements in waveform emulation techniques provide valuable insights and benchmarks that facilitate seamless connectivity transitions from Wi-Fi to LoRa networks, thereby enhancing interoperability and efficiency in heterogeneous wireless communication environments. The transition from Wi-Fi to LoRa shares similarities with earlier methodologies: as long as the waveforms of a signal generated by a Wi-Fi chip are sufficiently close to those produced by an IoT transmitter (e.g., LoRa), the corresponding IoT receivers are able to decode the signals correctly.

Currently, LoRa takes advantage against Wi-Fi thanks to its higher receiver sensitivity, capable of reaching -134 dBm, and its longer transmission range, exceeding 500 m [18]. Consequently, direct transmission of data from Wi-Fi to LoRa would improve the overall transmission capabilities of the network, thus providing a promising market opportunity. At the same time, according to a recent forecast by IDC [29], more than 4.1 billion Wi-Fi devices are supposed to be shipped only in 2024, with a total of around 19.5 billion devices currently being in use. The number of public LoRaWAN networks, which are available globally, has grown by 66% over the past three years. The installed base of LPWAN devices will exceed 2 billion units in 2025 [30].

TABLE I
KEY LoRa PHY PARAMETERS

Model Parameter	Symbol	Options
Frequency	f	2.4 GHz
Spreading factor	SF	5-12
Bandwidth	BW(KHz)	203, 406, 812, 1625
Code rate	CR	4/5, 4/6, 4/7, 4/8
Cyclic Redundancy Check	CRC	on or of

III. PRELIMINARY INFORMATION

Orthogonal Frequency Division Multiplexing (OFDM) constitutes a foundational component underpinning numerous contemporary modulation schemes, including 802.11 WLAN (Wireless Local Area Network), 802.16 WiMAX (World Interoperability for Microwave Access), and 3GPP (3rd Generation Partnership Project, Long Term Evolution) 5G. OFDM represents a digital multi-carrier modulation framework that expands upon the notion of single subcarrier modulation by concurrently employing numerous closely spaced orthogonal subcarriers within a solitary channel. In contrast to transmitting a high-rate data stream using a single subcarrier, OFDM harnesses a considerable array of orthogonal subcarriers, each subjected to conventional digital modulation schemes (e.g., QPSK:Quadrature Phase Shift Keying, 16-QAM:Quadrature Amplitude Modulation) at a reduced symbol rate. The merge of these numerous subcarriers culminates in data transmission rates akin to conventional single-carrier modulation schemes within equivalent BW constraints.

The accompanying diagram elucidates the fundamental principles of an OFDM signal and the interplay between the frequency and time domains. Within the frequency domain, multiple adjacent tones or subcarriers independently undergo modulation with intricate data. A subsequent Inverse Fast Fourier Transform (IFFT) transmutes the frequency-domain subcarriers into the temporal domain, thereby yielding the OFDM symbol. Subsequently, in the time domain, guard intervals are interjected between symbols to avert inter-symbol interference at the receiver, stemming from multi-path delay spread within the radio channel. Multiple symbols may be concatenated to construct the ultimate OFDM burst signal. At the receiver, a Fast Fourier Transform (FFT) operation is executed on the OFDM symbols to retrieve the original data bits.

Semtech's LoRa is an industry-leading LPWAN technology that encompasses various wireless technologies and facilitates the development of scalable IoT networks. Initially intended for sub-GHz band operation, LoRa technology has since 2017 incorporated 2.4GHz LoRa transceivers, such as SX1280/SX1281, which exhibit similar performance to their sub-GHz counterparts (i.e., 470-510MHz for China, 433/868 MHz for EU) [31]. Moreover, the SX1280 LoRa module has an extended communication range and is robust against interference in this commonly used band, as evidenced in Table. I. Furthermore, it can receive LoRa packets without stringent channel duty cycle restrictions. Concurrently, the maximum available BW has been augmented from 500kHz to 1625kHz, resulting in faster data rates (i.e., from 21kbps to 70kbps). The SX1280 offers lower SF options, with a minimum setting of SF=5. This means that the duration of

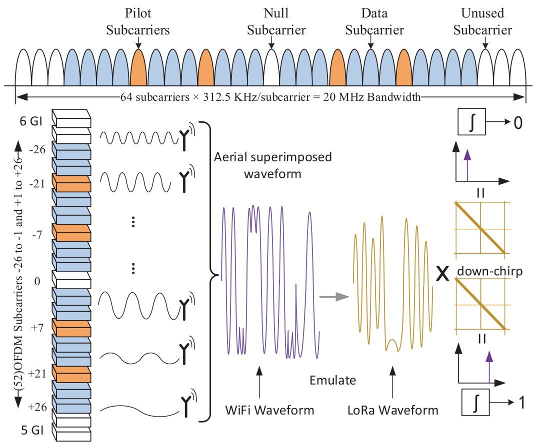


Fig. 2. We use Wi-Fi waveforms to emulate LoRa waveforms in time domain.

the LoRa chirp is further reduced under the same BW configuration. Meanwhile, the SX1280 supports four selectable coding rates (CR). For example, when CR=4/8, it means that an equal amount of redundancy is added to the data set to enhance its interference resistance. Therefore, we can dynamically select the appropriate CR based on environmental conditions. Consequently, the 2.4GHz LoRa technology has garnered significant attention and offers wider support for IoT applications, such as low-latency healthcare connections.

IV. WiLo in a Nutshell

Fig. 2 depicts the WiLo architecture and the data transmission process from a Wi-Fi to a LoRa device, which involves transmitting a Wi-Fi frame containing a chosen payload and generating a specific signal using OFDM modulation, resulting in a waveform that can be recognized by a LoRa receiver as a valid LoRa frame. The principle of WiLo: given that the waveforms produced by a Wi-Fi chip closely approximate those emanating from a LoRa transmitter, the corresponding LoRa receiver is poised to accurately decode the signals. The modulated signal is composed of single-tone sine waves. The payload of a Wi-Fi frame is constructed by WiLo and the payload is spread into chips and transmitted as a wide-band signal. After receiving this signal, the signal passes through a low pass filter and successfully triggers a standard LoRa RX procedure.

It should be noted that the content of the Wi-Fi preamble, header, and trailer cannot be controlled and therefore cannot be utilized to generate desired signals. These parts are treated as noise and ignored by the LoRa receiver. The LoRa device multiplies it with the down-chirp in the time domain (detail is shown in Section.V-B) and performs a FFT analysis, which yields a detected peak in the frequency domain that corresponds to the Wi-Fi signal (assuming perfect synchronization). By detecting this peak, the base station can identify and track the bit streams of Wi-Fi signal, decode them, and reassemble them into Wi-Fi frames. This decoding process enables the LoRa device to extract the original payload data transmitted by the Wi-Fi device.

The transmission, detection, and decoding of Wi-Fi signals facilitate efficient and dependable wireless communication from Wi-Fi to LoRa. The WiLo system has been designed based on two key observations to achieve long-range communication. Firstly, by manipulating the frame payload and modulation schemes of Wi-Fi, particularly using OFDM, specific signals can be generated that consist of continuous single-tone sinusoidal waves, which can be detected by a receiver. Secondly, the demodulation technique used in LoRa receivers based on FFT can detect and demodulate these specific signals from Wi-Fi.

V. WiLo Design

A. Primer of OFDM Modulation

The utilization of OFDM techniques is specifically employed in the IEEE 802.11a/g/n/ac Wi-Fi standards. For IEEE 802.11a/g/n, each channel in the 2.4 GHz frequency range occupies 20 MHz of bandwidth. Additionally, each channel is divided into 64 subcarriers, each with a bandwidth of 312.5 kHz. Of these, 52 are used as data subcarriers (48 data subcarriers in IEEE 802.11a/g), resulting in an effective bandwidth of 16.25 MHz for actual data transmission. Additionally, each sub-carrier has the capability to utilize a unique modulation scheme, such as QPSK or 16-QAM, depending on the characteristics of the physical channel being utilized in the Wi-Fi communication. This allows for adaptive modulation schemes to be employed based on the quality of the physical channel being used. This enhances the overall efficiency and performance of the Wi-Fi communication system.

The use of OFDM and the ability to adaptively select modulation schemes contribute to the robustness and reliability of Wi-Fi communication in various environments. These features are important factors in the successful implementation of Wi-Fi networks for modern wireless communication. Furthermore, several papers, among which [32], [33], demonstrate the successful deployment of IEEE 802.11a and IEEE 802.11g and more recently IEEE 802.11ax standards in industrial scenarios, for efficient and reliable wireless communication. Thus, understanding the technical details of these implementations is crucial for researchers and practitioners in the field of wireless communication. Overall, the use of academic language and precise terminology helps to accurately convey the technical intricacies of Wi-Fi communication systems.

A data source supplies the binary data vector b, while a mapper, such as QAM, converts the binary data into symbols from a complex constellation with 2^μ possible values, where μ represents the modulation order. The resulting vector d can then be decomposed into low symbol-rate streams consisting of K sub-carriers using an OFDM modulator. Overall, this process involves transforming binary data into complex symbols and organizing them into sub-carriers for transmission using OFDM modulation. This is a common technique used in communication systems for efficient data transmission. The use of precise academic language helps convey the technical details of the process. As shown in Fig. 3, the input of the OFDM modulator is given by,

$$d = [d_0, d_1, \ldots, d_{K-1}]^T \tag{1}$$

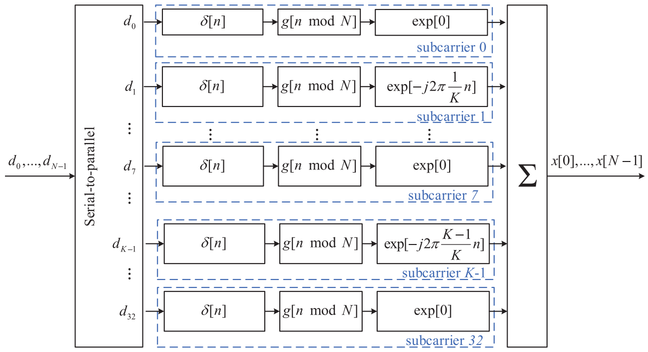


Fig. 3. Details of the OFDM modulator.

where, $(\cdot)^T$ is the transpose operator and the total number of symbols is $N = \mu K$. The transmitted signal at the discrete-time index n is given by,

$$x[n] = \sum_{k=0}^{K-1} d_k g_k[n], n = 0, 1, \ldots, K-1 \quad (2)$$

where, $g_k[n]$ is the corresponding pulse shaping filter, can be expressed as,

$$g_k[n] = g[(n\ mod\ N)]e^{\frac{-j2\pi kn}{K}} \quad (3)$$

with $g[n]$ representing the prototype filter and mod denoting the modulo N operation. The exponential function performs the frequency shifting operation and n is the sampling index. By collecting N samples of $g_k[n]$ in the matrix form, the vector representation of Eq. 2 can be written as,

$$x = Ad \quad (4)$$

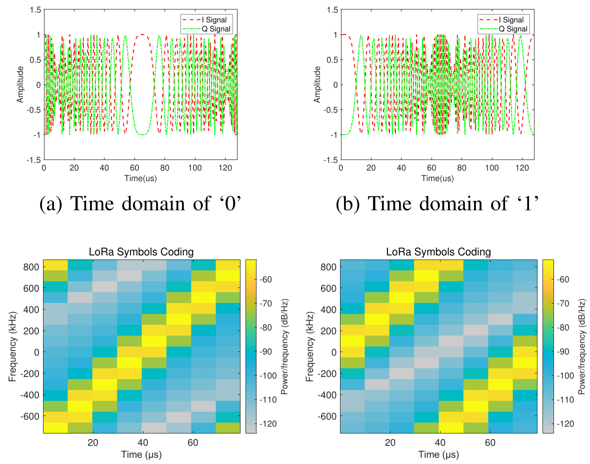
where, $x = [x[0], x[1], \ldots, x[N-1]]^T$, and $A = [g[0] \ldots g[K-1]]$. Before transmitting, the Cyclic Prefix (CP) of length N_{CP} is added to create the vector given as,

$$\hat{x} = [x(N - N_{CP} : N-1)^T, x^T]^T \quad (5)$$

The CP length is taken equal to the number of taps in the channel. OFDM has 48 data subcarriers and the OFDM bandwidth is 20 MHz. The LoRa bandwidth is much smaller and we only need to control some subcarriers to simulate the LoRa waveform, such as, $[x[n-k], x[n-k+1]], n = 1, 2, .., k = 1, 2, \ldots, k < n$. However, except for the subcarriers that need to be controlled, other subcarriers cannot be set to null, so all subcarriers need to be considered to meet the LoRa waveform requirements.

B. A Primer of Chirp Spread Spectrum (CSS) Modulation

The use of chirp pulses in LoRa modulation allows for efficient data transmission over long distances with low power consumption. Chirp signals are a type of continuous wave signal with a frequency that varies linearly over time. The linear variation in frequency results in a spread-spectrum signal that occupies a wide bandwidth. For LoRa, two different chirps are used to represent bit '0' and bit '1'. The spectrograms of these two chirps are shown in Fig. 4. These chirps are then transmitted using the frequency-shift keying (FSK) modulation [34], with the frequency of the transmitted signal alternating between the frequency of the up-chirp and the down-chirp.



(a) Time domain of '0' (b) Time domain of '1'

(c) Frequency domain of '0' (d) Frequency domain of '1'

Fig. 4. The spectrograms of bit '0' and bit '1' using FSK.

The Chirp Spread Spectrum (CSS) demodulation technique utilized by LoRa takes advantage of the fact that up-chirps with different initial frequencies produce cross-correlation peaks at different locations. This is illustrated in Fig. 4, where the cross-correlation peaks for two different up-chirps are shown. To demodulate the received chirps, the LoRa receiver multiplies them with the down-chirp in the time domain. This operation is equivalent to performing a convolution in the frequency domain. The down-chirp and the up-chirp are highly correlated due to their conjugation, resulting in a strong correlation peak at an FFT frequency bin, i.e., an interval between samples in the frequency domain. This peak corresponds to the frequency of the transmitted chirp and is used to recover the transmitted data.

In the LoRa demodulation process, the received signal is first multiplied with the down-chirp in the time domain. This is followed by an FFT operation that transforms the signal from the time domain to the frequency domain. The resulting spectrum obtained through FFT displays the frequency components of the received signal. When demodulating LoRa signals, the FFT spectrum shows a peak at the 0th bin for the demodulation of the chirp signal corresponding to bit '0'. However, the peak for the bit '1' is situated in the middle of the FFT bins. This difference in peak location arises from the initial frequency of the up-chirp used to represent bits '0' and '1', as is shown in Fig. 5.

The LoRa technology employs the CSS demodulation technique as a pivotal element in the retrieval of transmitted data, capitalizing on the nuanced distinctions in peak location within the received signal. More precisely, the CSS technique serves as a pivotal tool for peak localization within the FFT spectrum, thereby enabling the receiver to discern whether the received signal corresponds to a binary '0' or '1'. The incorporation of CSS demodulation in conjunction with chirp pulses within the LoRa modulation framework not only furnishes a robust and efficient data transmission mechanism but also does so with

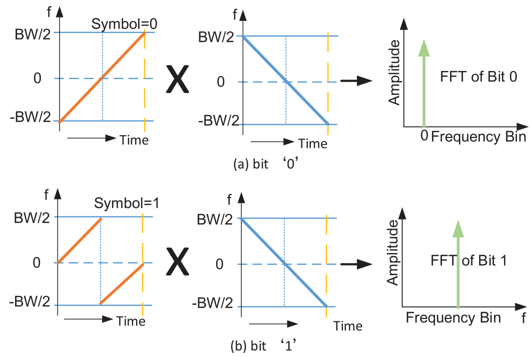


Fig. 5. The CSS modulation baseband implementation for LoRa to transmit '0'/'1'.

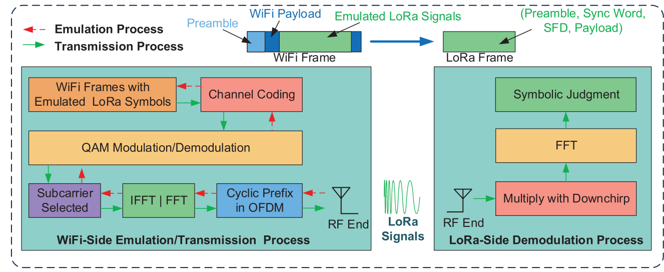


Fig. 6. The WiLo architecture employing OFDM to emulate the CSS.

a noteworthy emphasis on power efficiency, particularly when data needs to traverse considerable distances. In this context, LoRa stands out not only for its transmission reliability but also for its judicious management of power resources, ensuring that data can be transmitted over extended ranges while preserving low energy consumption.

C. LoRa Chirp Emulation With 802.11g OFDM

Due to the significant discrepancy between Wi-Fi bandwidth and transmission rate parameters in comparison to those of LoRa, we emulate the Wi-Fi-to-LoRa communication through a simulation approach. In the 2.4 GHz frequency spectrum, LoRa supports a maximum bandwidth of merely 1.625 MHz, whereas Wi-Fi boasts a substantially wider bandwidth, reaching 20 MHz, with newer 802.11 standards introducing broader bandwidth options such as 40 MHz, 80 MHz, and 160 MHz [2]. This article introduces the workflow for CTC communication at the LoRa physical layer. This framework can be dissected into two primary segments: one entails the simulation and transmission processes situated on the Wi-Fi side, while the other encompasses the demodulation process on the LoRa side, which remains transparent throughout the entire CTC operation on the LoRa side. Notably, no hardware or software modifications are necessitated on the LoRa side, which is chiefly responsible for generating the requisite effective load signal for simulation and reception evaluation.

At the heart of Wi-Fi operation lies the simulation process, which essentially seeks to establish a mapping relationship between the effective load of LoRa symbols and Wi-Fi frames. This entails transmitting the LoRa time-domain signal to the OFDM receiver, thereby executing the reverse simulation process within software implementation. The simulation process entails the selection of specific sub-carriers corresponding to the LoRa bandwidth and assigning weights. Subsequently, these selected subcarriers undergo demodulation. Since the LoRa signal does not employ QAM, the demodulation process aims to match the nearest QAM constellation point to the received signal, ensuring its proximity to the required LoRa signal. Finally, channel coding is employed to generate a Wi-Fi frame containing the LoRa symbol. Once the simulation concludes, a frame is produced, which LoRa can recognize by combining the simulation results.

In a broader context, a standard Wi-Fi frame comprises a preamble and a data payload. Given the mentioned substantial difference in bandwidth between Wi-Fi and LoRa, the data payload section of the Wi-Fi frame can accommodate an entire LoRa frame. This enables the selection of suitable central frequencies for both Wi-Fi and LoRa. Following the simulation process, when Wi-Fi transmits a Wi-Fi frame to a LoRa receiver, LoRa can identify the LoRa frame within the Wi-Fi frame while disregarding other irrelevant components. Ultimately, LoRa can successfully extract the LoRa frame integrated into the Wi-Fi frame, thereby realizing cross-technology communication from Wi-Fi to LoRa. Fig. 6 illustrates the WiLo architecture employing OFDM to emulate the CSS. Starting from the left side of Fig. 6, the pre-generated LoRa waveform, intended for emulation, undergoes the Emulation Process, resulting in a WiFi payload. This WiFi payload is then modulated into a WiFi waveform and transmitted. The LoRa receiver, upon performing physical layer demodulation, can successfully decode the CTC data transmitted by WiLo.

1) LoRa Symbols Are Aligned With Time/Frequency Domains of OFDM: In order to achieve signal emulation, it is crucial to consider the physical layer constraints inherent in both technologies. Specifically, when emulating a chirp for transmission via Wi-Fi, it is essential to ensure that the frequency range and time duration of the emulated chirp align with the characteristics of the target LoRa chirp. This necessitates careful consideration and synchronization of the frequency range and time duration parameters of the chirp emulation to accurately replicate the behavior of LoRa chirps on real Wi-Fi and LoRa devices.

The chirp waveform, a fundamental component within the LoRa modulation framework, is defined by a parameter known as the spreading factor (SF), which, central to the characteristics of the chirp waveform, governs the overall structure of the waveform. Specifically, the chirp waveform comprises a total of 2^{SF} individual samples, each sampled at a rate that aligns with the bandwidth associated with the signal. To provide an example, let us consider a scenario where the SF is configured at a value of 7, and the BW of the signal is set at 1625 kHz. In such a case, the temporal extent of the chirp waveform can be calculated as follows:

$$T_{chirp} = \frac{2^{SF}}{BW} = \frac{128}{1625KHz} \approx 79\mu s \qquad (6)$$

The Wi-Fi (802.11g) device operates at a chip rate of 54 Mbps, corresponding to a chip duration of $4\mu s$. In order

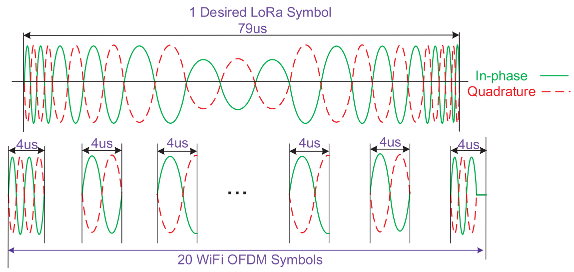


Fig. 7. Emulate CSS with Wi-Fi QAM.

to generate a waveform with a duration equivalent to that of a single LoRa chirp, it is necessary to transmit approximately 79/4 (approximately 20) chips. In the context of OFDM modulation in the 54 Mbps mode, each payload bit is modulated into one OFDM chip. Thus, it requires 20 bits to emulate a chirp waveform commonly used in LoRa communication. Consequently, a complete LoRa symbol needs to be segmented and emulated by 20 individual Wi-Fi symbols, as depicted in Fig. 7. Once identifying the sample segment, the LoRa device performs a multiplication operation with the down-chirp in the time domain, followed by a FFT analysis. WiLo, on the other hand, detects frequency peaks from the FFT output and proceeds to locate the preamble of a LoRa frame.

The preamble consists of a sequence of repeated peaks that serve as an indicator for the beginning of a frame, allowing WiLo to precisely identify the initiation of the data transmission. Once the preamble is detected, WiLo tracks the corresponding peaks and demodulates them in parallel into symbols or bits. This process involves the conversion of the analog signal into a digital signal, enabling subsequent processing and analysis by the receiving device. Finally, the demodulated bits are organized into frames, completing the demodulation process. By following these steps, WiLo successfully demodulates LoRa frames and extracts the pertinent information they encapsulate, thereby facilitating reliable and efficient communication between devices.

2) OFDM Symbols Emulate LoRa Signal: Achieving time domain signal alignment involves dividing a LoRa symbol signal into 20 LoRa sub-signals for separate emulation. Consequently, emulating one LoRa symbol yields 20 OFDM symbols. The following section provides a detailed account of the emulation for a single LoRa sub-signal. Initially, the time domain signal undergoes upsampling to ensure compatibility with the simulation, as all signals involved are complex and baseband signals. For the Wi-Fi physical layer, a sampling rate of 20 MHz satisfies the Nyquist sampling law. Subsequently, in accordance with OFDM modulation, the first 1/5 (i.e., the initial 0.8 μs) of the received signal is discarded, representing the CP segment. In other words, for a 4 μs LoRa sub-signal, only the subsequent 3.2 μs is effectively emulated.

Utilizing the FFT algorithm, computational operations are performed on the 3.2 μs time domain signal, resulting in the frequency domain component corresponding to its specific bandwidth. In the case of 802.11a/g, 64 FFT points represent 64 subcarriers. Assuming the original signal is denoted as X, the frequency domain component after the FFT operation can be expressed as follows:

$$F(n) = \sum_{j=1}^{64} X(j) e^{-2\pi i \frac{(n-1)(j-1)}{64}} \tag{7}$$

According to Eq.7, it is straightforward to determine the value range of n in the frequency domain component $F(n)$, which spans from 1 to 64. This range represents the frequency domain component associated with each OFDM subcarrier. In the equation, j denotes the sampling point of the time domain signal X, while i represents the imaginary unit. By analyzing the frequency domain component, we can obtain both the frequency values of the frequency points and the amplitude characteristics of the frequencies. The amplitude A for each frequency component can be calculated as $\frac{abs(F(n))*2}{N}$, where N is equal to 6. The term abs indicates the magnitude of the complex number. Assuming that the sampling rate of the current signal is Fs, the frequency corresponding to the $n-th$ frequency domain component can be expressed as $((n-1)*Fs)/N$, where $N = 64$.

Once completion of the FFT operation, the frequency domain components corresponding to each subcarrier can be obtained. As mentioned in the previous section, emulating a LoRa signal requires the utilization of 6 OFDM subcarriers (although 7 subcarriers may be necessary if the center frequency of LoRa is adjusted accordingly). Consequently, the subsequent step involves the careful selection of the required subcarriers, specifically the FFT bin. Since there exists a mapping relationship between the FFT bin and the actual OFDM subcarriers, a straightforward approach to selecting the appropriate FFT bin involves calculating the average result of the FFT operation across all symbols. Subsequently, a histogram is generated, which serves as a reference for identifying the OFDM subcarriers to be utilized by the WiLo system. Subsequently, QAM decoding is performed on the segment of the FFT result that corresponds to the data subcarrier.

In the QAM decoding process, since the LoRa signal does not undergo QAM coding, some frequency components may deviate from the standard constellation points. To address this, we employ a technique called quantization, as utilized in WEBee [26] and WIDE [35], where the closest constellation points are selected to replace these deviated frequency components. Following the QAM decoding, it is necessary to set the constellation points, other than those corresponding to the valid subcarriers, to zero. This step is undertaken to minimize the interference caused by the subcarriers during the emulation process. Upon completion of QAM decoding, the inverse process of channel coding is performed. Channel coding employs convolutional codes, and the decoding process utilizes the decoding algorithm to decode the QAM decoded symbols into bit data, which is subsequently stored for further analysis and processing, as is shown in Fig. 8.

In the process of emulating a LoRa sub-signal, a coherent set of binary data is generated, representing the underlying information carried by that sub-signal. The emulation procedure extends beyond the generation of a singular sub-signal, as it is designed to replicate the transmission of LoRa symbols comprehensively. By cyclically emulating the sub-signals

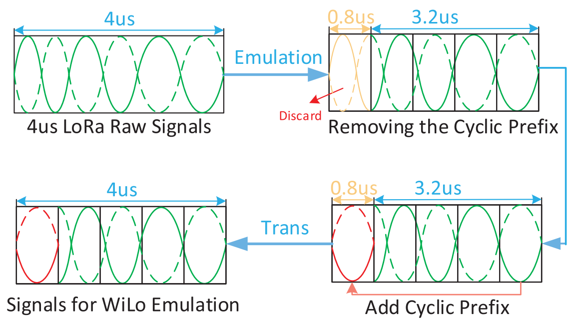


Fig. 8. Emulate CSS with Wi-Fi QAM.

associated with a LoRa symbol, a discrete set of N individual bit data sequences is produced. The value of N is contingent on the temporal relationship between the duration of a LoRa symbol and that of an OFDM symbol, precisely defined as $(N = (time_{LoRasymbol})/(time_{OFDMsymbol}))$. These N distinct sets of binary data, originating from the emulation of multiple LoRa sub-signals, collectively assemble into a coherent string of bit data. This unified bit data sequence faithfully mirrors the content of a single LoRa symbol. It is within this assemblage of bit data that the crucial information and encoded messages of a LoRa symbol are encapsulated, embodying the essence of LoRa signal transmission and facilitating its comprehension and utilization in communication systems.

3) WiLo Generates LoRa Signals and the Transmission Process: Once the successful emulation of LoRa symbols is accomplished, a sequence of binary data, denoted as the source data, is acquired. Subsequently, the source data undergoes modulation and is transmitted to the LoRa receiver through the utilization of the OFDM modulation technique. This process unfolds through several phases. Initially, the source binary data is subject to convolutional encoding, augmenting its error correction capabilities. Following convolutional encoding, QAM encoding is applied. In this phase, groups of 6 bits are considered, and 64-QAM encoding is executed to yield a 64-QAM symbol.

Subsequent to this stage, the simultaneous allocation of multiple QAM symbols are carried onto the data subcarrier. This operation permits the amalgamation of multiple QAM symbols onto the data subcarrier during the process of OFDM modulation. Following this, the frequency domain component is converted into a time domain signal via an inverse Fourier transform (IFFT). This transformation effectuates the transition from the frequency domain representation to the time domain.

To ensure the provision of a guard interval and the mitigation of inter-symbol interference, a cyclic prefix of 0.8 μs is appended to each OFDM symbol in the time domain signal. This cyclic prefix serves as a guard interval, effectively separating consecutive symbols. By executing the aforementioned steps, a sequence of OFDM signals, encompassing the emulated LoRa signals, is generated, primed for transmission through the front-end Radio Frequency (RF) module to reach the LoRa receiver.

4) How to Control OFDM Subcarriers to Generate Specific LoRa Signals: To control specific subcarriers and generate the desired LoRa time domain signal, WiLo employs a process that involves careful selection of QAM constellation points within certain constraints. This process aims to approximate the desired LoRa time domain signal by manipulating the subcarriers in the frequency domain. Let $X[n]$ represent the OFDM encoded symbol, where OFDM utilizes 64 subcarriers, including the null subcarrier. The IFFT equation is utilized to determine the time domain signal, given by

$$S(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{2\pi i\frac{nk}{N}} \tag{8}$$

where $X(k)$ is the frequency domain component on each subcarrier, k denotes the k-th subcarrier, n represents the sampling points of the generated time domain signal, and N represents the length of the IFFT. Eq. 8 can be rewritten as,

$$S(n) = \frac{1}{N}[X(0) * e^{2\pi i\frac{0*k}{N}} + X(N-1) * e^{2\pi i\frac{(N-1)*k}{N}}] \tag{9}$$

The summation in Eq. 9 indicates that the time domain signals of all subcarriers are superimposed to obtain the complete time domain signal of OFDM. The n denotes the sampling points on the time-domain signal. WiLo uses 6 data subcarriers to communicate with a LoRa, and assuming that WiLo uses the first 6 subcarriers, i.e., $k \in [0, 5]$, the time domain signal corresponding to the selected subcarriers, $S_{sub}(n)$, can be derived according to Eq. 9 as,

$$S_{sub}(n) = \frac{1}{N}[X(0) * e^{2\pi i\frac{0*k}{N}} + X(5) * e^{2\pi i\frac{4*k}{N}}] \tag{10}$$

In Eq. 10, the variable N represents the number of subcarriers, which is different from N that corresponds to the 6 subcarriers used in the specific OFDM symbol. The variable n still denotes the sampling points on the time-domain signal. It is important to note that Equation (1)-(3) only represents the time domain signal of a single OFDM symbol with 6 subcarriers, spanning a duration of 4 microseconds. However, as per the previous configuration, we require 20 OFDM symbols to form a complete LoRa symbol. Hence, we need to concatenate 20 time domain signals of OFDM symbols with 6 subcarriers each. The resulting time domain signal, denoted as $S_{Lora}(m)$, is given by $S_{sub1}(n), S_{sub2}(n), S_{sub3}(n), \ldots, S_{sub19}(n), S_{sub20}(n)$.

Here, $n \in N$ represents the sampling point of the time domain signal for each OFDM symbol, with its value determined by the sampling rate. On the other hand, $m \in M$ represents the sampling points of the time domain signal required for the final generation of the LoRa signal, where M should be equal to $M = 20 * N$. The final LoRa receiver will demodulate the time domain signal represented by $S_{Lora}(m)$ to extract and decode the CTC data transmitted by WiLo. This process enables the receiver to retrieve the intended information carried by the WiLo-generated LoRa signal.

D. Multi-Channel Communication

The WiLo proposed in this paper has the potential capability in parallel communication. Wi-Fi adopts OFDM modulation

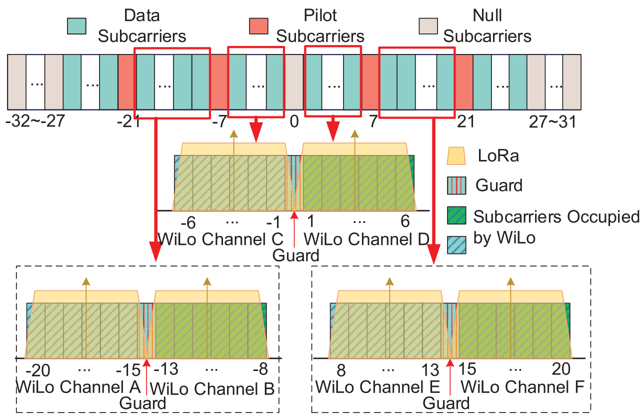


Fig. 9. Wi-Fi Subcarrier Allocation for LoRa Signal Emulation using QAM.

and divides the channel BW into 64 subcarriers, which contain 48 data subcarriers as well as 4 guide frequency subcarriers, and the BW of each subcarrier is 312.5 kHz (20 MHz/64), so the actual BW occupied by the data subcarriers is 15 MHz (312.5 kHz*48). Assuming that the BW of LoRa is set at 1.625 MHz and each LoRa corresponds to 6 subcarriers, WiLo can theoretically realize up to 8 LoRa parallel communications. However, in practice, since the data subcarriers are often not continuous, and there will be frequency-conducting subcarriers and null subcarriers in between, and WiLo must select continuous available subcarriers to emulate LoRa signals, the number of WiLo parallel communication is less than the theoretical maximum of 8.

The distribution of Wi-Fi subcarriers is shown in Fig. 9, which illustrates 4 available data subcarrier intervals and 2 unavailable data subcarrier intervals. The 2 unavailable subcarrier intervals are unavailable because their contiguous BW is too narrow to meet the demand of LoRa signal emulation. The subcarrier intervals $8 \sim 20$ and $-20 \sim -8$ satisfy the BW requirement, and assuming that the center frequency of LoRa can be adjusted arbitrarily, this interval can support parallel communication with 2 LoRa devices respectively, and the subcarrier interval $1 \sim 6$ and the subcarrier interval $1 \sim 6$ can also communicate with 1 LoRa device respectively. Additionally, WiLo supports the emulation of LoRa signals with various BW configurations. Smaller BW configurations can effectively increase the number of parallel communications, but they also reduce the resilience of the emulated signals against interference. In practical scenarios, WiLo can dynamically adjust the LoRa BW configuration based on the level of environmental interference, enabling efficient CTC communication. While the analysis in Fig. 9 assumes that each WiLo channel operates with the same BW, in practice, WiLo allows for parallel transmissions using different BW configurations. WiLo can allocate varying numbers of OFDM subcarriers to different emulated channels, facilitating more complex parallel communication strategies.

VI. Performance Evaluation

We now present the obtained empirical results and a detailed account of the experimental setup.

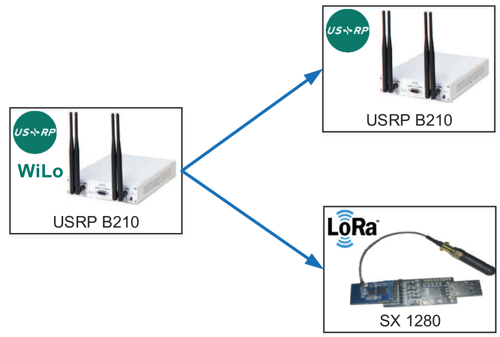


Fig. 10. Experimental setup.

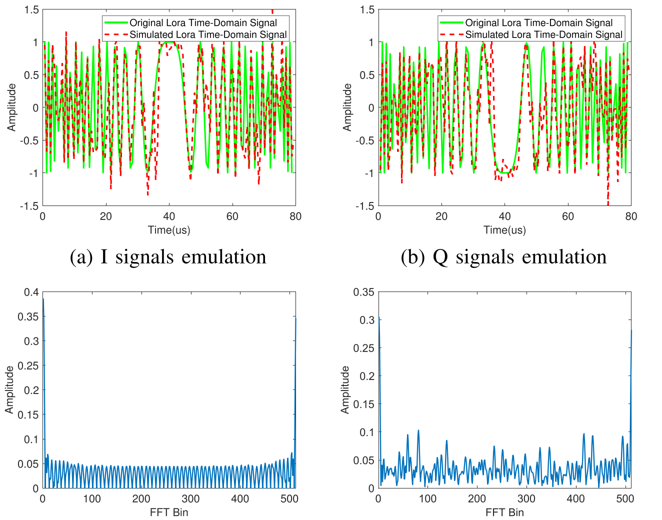
A. Hardware

The experimental setup of the WiLo system is depicted in Fig. 10. The experimental evaluation unit includes the Universal Software Radio Peripheral (USRP)-B210 platform with LoRa physical layer (PHY), a commercial LoRa device based on the SX1280 chip. The system consists of 2 types of devices: (i) the sender, which uses the USRP-B210 platform with 802.11 b/g PHY, and (ii) the receiver, which uses a commodity LoRa receiver (i.e., Semtech SX1280) and the USRP-B210 platform with a LoRa PHY. We would like to emphasize that WiLo is directly supported among commodity devices, and the USRP-B210 devices are used only for evaluation purposes to measure low-level PHY information, which is inaccessible by commodity devices. For example, a commodity Wi-Fi card such as the Atheros AR2425 can replace USRP-B210 devices as the sender. The WiLo receiver (i.e., LoRa receiver, Semtech SX1280 chip) uses a BW of 1.625 MHz and a spreading factor of 7, with the channel frequency set at 2.4 GHz.

B. Configuration of WiFi Signals

We have successfully implemented the WiLo emulation method and have conducted a comparative study to evaluate its impact on LoRa signals. Fig. 11 and Fig. 12 plot the specific Wi-Fi waveforms utilized for emulating LoRa chirps in WiLo for bit '0' and '1'. In particular, we focus on the emulation effect of LoRa symbol 0, which is taken as a case study. Our results, illustrated in Fig. 11a and Fig. 11b, demonstrate the emulation effect of the I-way and Q-way signals, respectively. The solid line represents the original LoRa time-domain signal, while the dashed line represents the WiLo-emulated LoRa time-domain signal. The WiLo-emulated LoRa signal closely approximates the original LoRa signal, which is essential given that LoRa utilizes the CSS modulation technique to transmit symbols by utilizing the initial frequency of chirp. LoRa signals are robust to noise, as each symbol utilizes the entire BW for transmission. Even if there is a deviation in the amplitude of the time-domain signal, LoRa can still correctly demodulate the symbol.

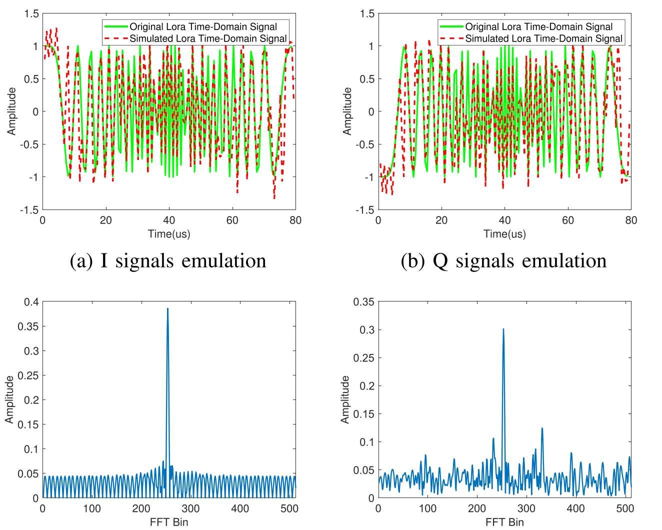
In the context of LoRa demodulation, the received signal is first multiplied with the standard downchirp signal, and the FFT operation is performed to determine the valid LoRa symbol based on the peak of the FFT operation result. In the absence of noise, the horizontal coordinate of the peak corresponds to the initial frequency of the chirp. The spectrum



(a) I signals emulation (b) Q signals emulation

(c) Native FFT for bit '0' (d) Emulation FFT for bit '0'

Fig. 11. Specific Wi-Fi waveforms utilized for emulating LoRa chirps in WiLo for bit '0'.



(a) I signals emulation (b) Q signals emulation

(c) Native FFT for bit '1' (d) Emulation FFT for bit '1'

Fig. 12. Specific Wi-Fi waveforms utilized for emulating LoRa chirps in WiLo for bit '1'.

plot of the original LoRa symbol 0 after FFT operation is illustrated in Fig. 11c, which indicates that the signal is free from noise interference. The peak appears near the starting position of the horizontal coordinate, and no significant peaks are generated at other positions. In contrast, the spectrum of the LoRa signal generated by WiLo after FFT operation is shown in Fig. 11d, where many spurious peaks are generated due to the simulation process error. However, the position of the maximum peak can still be distinguished relatively clearly, indicating that the corresponding symbol can be demodulated successfully by LoRa.

Therefore, the feasibility and emulation performance of WiLo are demonstrated, and the WiLo emulated LoRa signal can be correctly demodulated by the LoRa device. This finding is important because LoRa is known for its robustness to noise, even if there is an error in the amplitude of the time-domain signal. However, accurate simulation of LoRa signals is challenging due to the complex modulation techniques involved, and WiLo provides a viable solution for the accurate emulation of LoRa signals. The WiLo implementation allows for a controlled and systematic comparison of the WiLo emulated LoRa signal with the original LoRa signal. The choice of LoRa symbol 0 as a case study was appropriate since it represents the initial state of the signal and is an essential component of the signal demodulation process. The close resemblance of the emulated LoRa signal to the original LoRa signal, as shown in Fig. 11a and Fig. 11b, as well as in 12a and 12b, suggests that WiLo is a reliable method for accurate emulation of LoRa signals.

C. Evaluation of SF Configuration

Fig.13a depicts the Symbol Error Rate (SER) performance of WiLo for a BW of 1625 kHz under different SFs and Signal-to-Noise Ratios (SNRs). The results show that WiLo outperforms at higher SFs due to the longer chirp length that leads to more concentrated signal energy in demodulation. The ability of WiLo to concentrate signal energy enables it to achieve longer-range transmissions and higher noise resistance even in the presence of severe signal attenuation during propagation.

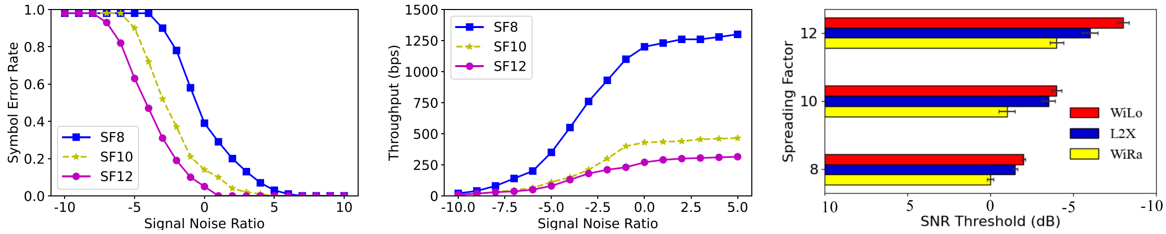
Furthermore, Fig. 13b illustrates the throughput of WiLo for different SF configurations. WiLo exhibits higher data rates for small SFs as they lead to shorter channel time for each chirp, allowing WiLo to encode more bits per chirp and experience lower SERs. Additionally, Fig. 13c presents the SNR thresholds of WiLo, L2X [36], and WiRa [37]. All these systems require SNR higher than 0 dB. WiLo's chirp spreading technique enables it to achieve an SNR threshold of -8 dB in SF=12, which is significantly better than L2X and WiRa, with thresholds of -6 dB and -4 dB, respectively.

D. Parallel Communication

In this study, we demonstrate that WiLo can enable 2 concurrent communication streams from Wi-Fi to LoRa. WiLo achieves this by utilizing channel mapping to transmit 2 LoRa frames, each occupying a distinct subcarrier region consisting of 11 data subcarriers within the Wi-Fi band. To assess the performance of this technique, we conducted experiments to measure the Frame Reception Rates (FRRs) of WiLo under different transmission distances for the left and right LoRa channels. As shown in Fig. 14, our results indicate that WiLo achieves FRRs above 96% for both parallel LoRa channels. However, we observe that the FRR for one channel is slightly lower than the FRR for the other channel. We attribute this difference in performance to the diversity of channel quality between the 2 channels, which can result in varying levels of interference or noise.

E. Indoor Scenario

To verify the soundness of the WiLo system, we chose an indoor environment as the site for experimental evaluation. As shown in Fig. 15, we have designed 2 indoor scenarios, i)



(a) Symbol Error Rate (b) Throughput (c) SNR threshold

Fig. 13. The performance of WiLo, L2X, and WiRa was compared under different SF configurations and varying SNRs.

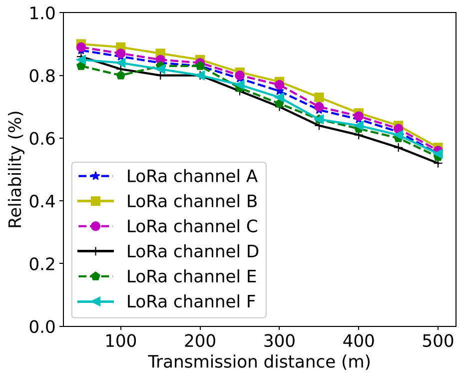


Fig. 14. Emulate CSS with WiFi QAM.

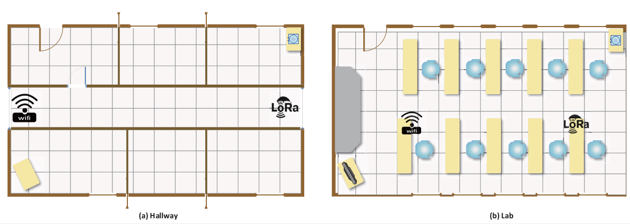
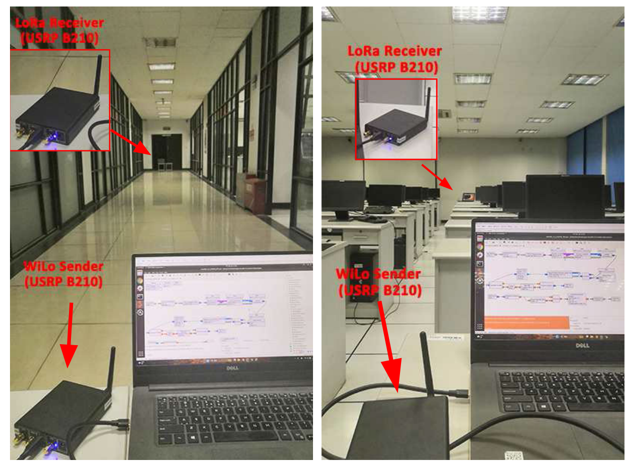
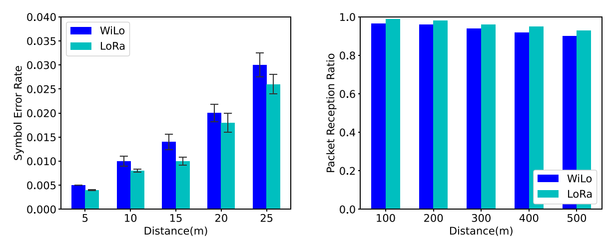


Fig. 15. Indoor Scene Plan.



(a) Hallway (b) Lab

Fig. 16. Indoor: Experimental setup.



(a) SER (b) FRR

Fig. 17. SER and PRR of WiLo of the indoor scenarios.

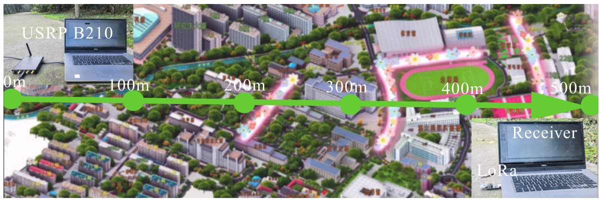
in the hallway and, ii) in the lab, and the used setups for those two sites are shown in Fig. 16. Regarding the detailed part of the experiment, the USRP B210 platform is used for the Wi-Fi transmitter and the USRP B210 platform with LoRa PHY is used for the LoRa receiver. For the parameter configuration, the LoRa SF is set to 7 and the BW is set to 1625 KHz.

We measure the frame FRR of the LoRa receiver by adjusting the transmitting distance at the WiLo transmitter from 5 m to 25 m in the indoor scenario. The experimental results are shown in Fig. 17. The figure shows that the SER has an increasing trend as the transmitting distance increases and the SER is lower than 0.04. Overall, the FRR is more than 96% overall in indoor environments, value that can be improved by continuously repeating the transmission.

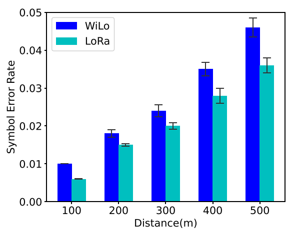
F. Outdoor Scenario

To evaluate the performance of WiLo in real-world outdoor environments, we conducted experiments along a campus road where we varied the distance between the sender and the receiver to 5 different locations to collect Wi-Fi packets from 100 m to 500 m. The bird view of the outdoor testbed is shown in Fig. 18a. The experimental settings were chosen meticulously to ensure the accuracy of the results, including the use of a SF and BW of 8 and 1625 kHz, respectively, and a transmission power of 20 dBm. The results of our experiment are compared to WiLo and commodity LoRa by measuring their SERs and PRR, as shown in Fig. 18b and Fig. 18c.

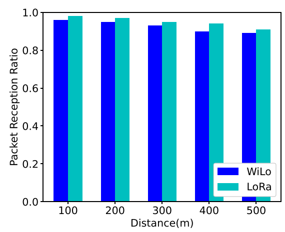
As the distance between sender and receiver increases from 100 m to 500 m, WiLo's SER also increases from 0.01 to 0.046. Similarly, commodity LoRa's SER increases from 0.006 to 0.036. Despite WiLo having a higher SER due to the imperfect emulated signal, it still manages to achieve a comparable performance to commodity LoRa in terms of both SER and PRR. This suggests that WiLo is an effective solution for achieving reliable and efficient wireless communication over long distances, which is particularly relevant to several applications and verticals such as remote monitoring systems, sensor networks, and smart cities.



(a) Setting



(b) SER



(c) PRR

Fig. 18. SER and PRR of WiLo of the outdoor campus-scale deployment.

VII. Disscussion

A. Potential Application Scenarios for the WiLo Framework

1) Smart Agriculture: WiLo can connect sensors across vast farmlands, enabling real-time monitoring of crop conditions, weather, and soil moisture. It covers long distances without the need for extensive infrastructure.

2) Smart Cities: WiLo can support the deployment of sensor networks for various smart city initiatives, such as air quality monitoring, traffic flow management, and public safety. The integration of Wi-Fi and LoRa technologies allows for flexible and cost-effective infrastructure development.

B. Differences Between the WiLo Framework and Heterogeneous Network Gateways

1) Cost-Effectiveness: WiLo leverages existing Wi-Fi infrastructure, reducing the cost of deploying and maintaining multiple gateways. Traditional gateways require configuration and maintenance for different IoT devices.

2) Flexibility: Unlike fixed-function gateways, WiLo employs a software-defined approach that can be updated or adjusted to accommodate new communication protocols or technologies.

C. Scalability and Dynamic Adaptability of WiLo

1) Scalability of WiLo: WiLo has been designed to operate efficiently across a range of devices and network configurations, making it a versatile solution for various IoT applications. The use of signal emulation techniques enables WiLo to function without requiring modifications to existing hardware, which significantly enhances its scalability. By leveraging standard Wi-Fi hardware to generate LoRa-compatible signals, WiLo can be deployed across a broad spectrum of devices, from consumer-grade Wi-Fi routers to industrial IoT gateways.

2) Adaptability to Different IoT Environments: WiLo can enhance industrial IoT applications by enabling seamless communication between various sensors and control systems in manufacturing plants. The system's ability to operate in challenging environments with high interference levels makes it suitable for industrial use cases. For smart city applications, WiLo can support the deployment of sensor networks for monitoring air quality, traffic flow, and public safety. The integration of Wi-Fi and LoRa technologies allows for flexible and cost-effective infrastructure development.

D. Potential Deployment Challenges

1) Security Concerns: Ensuring the security of data transmitted across WiLo is vital. The cross-technology communication must incorporate robust encryption and authentication mechanisms to protect against potential security threats.

2) Additional Power Consumption Due to CTC: The WiLo system, operating over WiFi, often results in additional energy consumption to simultaneously support CTC communication and WiFi transmission. Therefore, balancing the energy demands of WiFi transmission with the low-power characteristics of WiLo devices is crucial, especially in battery-powered IoT applications.

E. Future Research Directions

While WiLo shows great promise, several areas can be explored for further improvement. Future research could focus on optimizing data rates to better align Wi-Fi transmissions with the low data rates of LoRa. Adaptive data rate algorithms could be developed to dynamically adjust transmission parameters based on the network conditions and application requirements. Enhancing the energy efficiency of WiLo is crucial, especially for battery-powered IoT devices. Investigating power-saving mechanisms and optimizing the duty cycle of Wi-Fi transmissions can help minimize power consumption while maintaining communication reliability. Meanwhile, WiLo presents opportunities for integration with cutting-edge technologies such as large-scale models and 5G. For example, 5G can be utilized to perform software upgrades for WiLo on edge IoT gateways, while powerful cloud-based large models can leverage 5G to remotely manage and optimize WiLo operations.

VIII. Related Work

CTC [38] technologies open a new direction which enables direct communication among heterogeneous wireless devices through signal emulation, like WiZig [39] and ZigFi [40], despite their incompatible PHY modulation. WEBee [26] is a software-based solution for PHY-CTC that enables direct communication from Wi-Fi to ZigBee. It achieves this by modifying the Wi-Fi transmitter to emulate the ZigBee time-domain waveform, with the Wi-Fi device selecting the payload of a Wi-Fi frame to simulate the ZigBee packet. QAM emulation is at the core of WEBee, which is just one example of several PHY-CTC solutions. XBee [41] is a PHY-CTC from ZigBee to BLE, which interprets a ZigBee frame by observing the bit pattern obtained at the BLE receiver. LEGO-Fi [42] delivers information from ZigBee to Wi.Fi, besides many others such as Wi-Fi-to-LoRa [37], [43], Bluetooth-to-LoRa [44], [45], Bluetooth-to-LoRa [46] and LoRa-to-ZigBee [47], [48].

PMC [49] is waveform-based CTC from WiFi to ZigBee and uses the signals of the overlapping Wi-Fi subcarriers to emulate ZigBee signals, where the other subcarriers still transmit Wi-Fi signals. PMC firstly develops an offline search algorithm and maps the desired ZigBee signals to Wi-Fi QAM-modulated signals. LTE2B [50] focus on the wide-area network (e.g., LTE and Multefire) and is a representative CTC work that delivers information from LTE to ZigBee. WIDE [35] achieves CTC based on the method of digital emulation to reduce emulation errors, where the sender emulates the phase shifts associated with the desired signals instead of emulating the original time-domain waveform of the receiver. Authors in [9] comprehensively summarizes the CTC techniques and reveals that the key challenge for CTC lies in the heterogeneity of IoT devices, including the incompatibility of technical standards and the asymmetry of connection capability.

WiRa [37] firstly emulates LoRa waveforms with IEEE 802.11ax to achieve an efficient CTC from Wi-Fi to LoRa, where resource unit is used to emulate LoRa chirps and other resource units are set for high rate Wi-Fi users. In contrast to previous studies on CTC among IoT devices, this paper investigates the characteristics of LoRa in the 2.4 GHz bands and OFDM in 802.11g. To our knowledge, it represents the first systematic study on CTC from Wi-Fi to LoRa base on OFDM in 802.11g. WiLo, which enables Wi-Fi devices to establish long-range connections with a LoRa device, represents the initial framework towards a universal LPWANs. WiLo leverages the physical features of concurrent Wi-Fi transmission for decoding Wi-Fi packets at LoRa receivers and thus achieving reliable long-rang CTC. Our study is thus both complementary and orthogonal to prior works in the field.

IX. Conclusion

In this study, we propose WiLo, a novel approach that focuses on establishing direct communication from Wi-Fi to LoRa. Our work involves a comprehensive analysis of the characteristics of both Wi-Fi and LoRa networks. Through empirical investigations, we examin the characteristics of LoRa communication from a CTC perspective and derive insights to guide the design of WiLo. WiLo enables long-range CTC from Wi-Fi devices to LoRa devices. Extensive experiments were conducted to evaluate WiLo's performance, and the results demonstrate that it can reliably transmit Wi-Fi communication to LoRa over a distance exceeding 500 m, which significantly exceeds the range of native Wi-Fi communication. Therefore, WiLo has the potential to address applications requiring long-range communication by extending the communication range from Wi-Fi to LoRa networks.

Acknowledgment

The authors are grateful to the anonymous reviewers for their constructive comments which helped improve the quality of this article.

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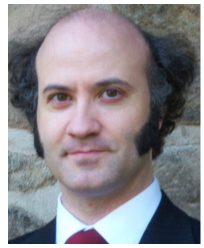

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